INCH-POUND
MIL-PRF-3098/33J
19 December 2003
SUPERSEDING
MIL-PRF-3098/33H
27 August 1997

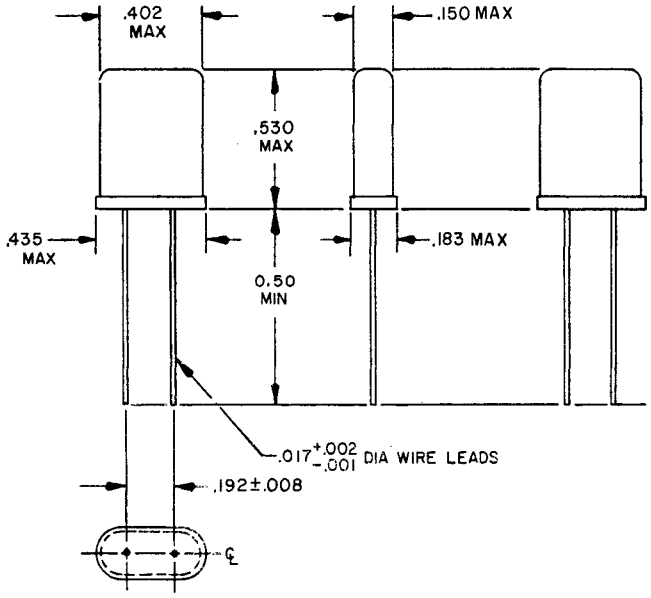
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR55/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 17 MHz to 62 MHz; third mechanical overtone; noncontrolled; series resonance.



Inches	mm	Inches	mm	Inches	mm
.001	.03	.150	3.81	.435	11.05
.002	.05	.183	4.65	.50	12.7
.008	.20	.192	4.88	.530	13.46
.017	.43	.402	10.21		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR55/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 17 MHz to 62 MHz, inclusive.

Capacitance, shunt: 7 pF, maximum.

Frequency tolerance (operating temperature range): ±50 parts per million (ppm).

Equivalent resistance: 40 ohms, maximum.

Mode of oscillation: Third mechanical overtone.

Calibration values:

Resistance: 40 ohms.

Resistor voltage drop: 0.28 volt.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Operating temperature range: -55°C to +105°C, inclusive.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration high frequency: Method 204 of MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army - CR
Navy - EC
Air force - 11
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 19, 99

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-0749-13)

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